7 **(a)** State what is meant by the binding energy of a nucleus.

.....

..... [1]

(b) The nuclear binding energy per nucleon varies with nucleon number.

On Fig. 7.1, draw a graph showing this variation for nuclei with nucleon number from 1 to 250. Label the axes with appropriate values.



Fig. 7.1 [3]

(c) Use your answer in **(b)** to explain why the fission of a nuclide having a high nucleon number is associated with a release of energy.

.....

.....

.....

.....

..... [2]

(d) A fissile material is an isotope capable of sustaining a nuclear fission chain reaction when struck by neutrons. Thorium-232 ($^{232}_{90}$Th) is used in certain nuclear reactors. Thorium-232 absorbs a neutron and then undergoes radioactive decay to form fissile uranium-233 ($^{233}_{92}$U).

(i) State the number of α-particles and β-particles emitted in this radioactive decay.

number of α-particles = ...

number of β-particles = ... [1]

(ii) Explain why it is not feasible for thorium-232 to absorb a helium-3 ($^{3}_{2}\text{He}$) nuclide to form fissile uranium-235 ($^{235}_{92}\text{U}$).

.....

.....

..... [1]

(e) When a nucleus of uranium-233 absorbs a neutron in a nuclear reactor, it can undergo fission to form various products and release further neutrons. In one such fission, xenon-137 and strontium-94 are produced together with the release of three neutrons. The reaction is shown in the equation.

$$^{233}_{92}\text{U} + ^{1}_{0}\text{n} \rightarrow ^{137}_{54}\text{Xe} + ^{94}_{38}\text{Sr} + 3\,^{1}_{0}\text{n}$$

Data for the nuclei in the reaction are given in Fig. 7.2.

nucleus	mass / u
neutron $^{1}_{0}\text{n}$	1.008664
uranium $^{233}_{92}\text{U}$	233.039634
xenon $^{137}_{54}\text{Xe}$	136.911557
strontium $^{94}_{38}\text{Sr}$	93.915355

Fig. 7.2

Use data from Fig. 7.2 to determine, to three significant figures, the energy associated with the change in mass in this reaction.

energy = ... J [3]

[Total: 11]